IMANUEL M.T. MANULLANG

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PROFILE

Information Technology student with hands-on experience across IT support, app security and development, including technical troubleshooting, vulnerability assessment, penetration testing, and backend development. Experienced in identifying and validating security vulnerabilities through Capture the Flag (CTF) competitions, penetration testing projects, and security labs using Burp Suite, Wireshark, Python and Linux. Certified in Google Cybersecurity (Coursera) and Cisco Ethical Hacker. Strong analytical and problem-solving skills with the ability to approach security from both attacker and developer perspectives.

SKILLS

Programming: Python, PHP, JavaScript, C++, SQL

App & Web Development: Laravel, WordPress (ACF), MySQL, PostgreSQL, Rest API

Security Tools: Burp Suite, OWASP/CWE, CVSS 3.1 scoring, Ghidra, GDB

Networking & Systems: Wireshark, Cisco Packet Tracer, Software-Defined Networking, Linux, Windows

Languages: Indonesian (Native), English (Intermediate)

CERTIFICATIONS & TRAINING

- Google Cybersecurity Specialization — Coursera
- Ethical Hacker — Cisco
- Cyber Sentinel Secure — X-code
- Security Manpower Training Program (SMTP) Indonesia — KISIA & NSHC Inc.
- Android Application Security Fundamentals — Mobile Hacking Lab

EXPERIENCE

Coding Instructor - UOB My Digital Space *Jul 2025 – Jul 2026*

Program by Ruang Guru x UOB

- Delivered a year-long coding curriculum in Scratch Jr for elementary students and Python for senior high students to 120 students across 4 partner schools, completing all 48 planned sessions on schedule.
- Built strong working relationships with each school's coordinator (PIC) to keep the program running smoothly across multiple locations and schedules.
- Mentored 43 students to the semifinal round and 3 students to the final round of the UOB My Digital Space Codefest 2026, national coding competition.

Freelance WordPress Developer - Property Listing Website *Feb 2026 – Jun 2026*

Azeroth Property Management (azerothpm.com)

- Designed and built a custom WordPress property management system over 4 months, from architecture and content structure to deployment.
- Implemented ACF-based content management and automated listing logic, supporting 80+ live property listings.
- Delivered the website based on the client's requirements and received positive feedback on the final result.

IT Support Intern *Mar 2025 – Jun 2025*

Kanwil Ditjen Perbendaharaan Provinsi Sumatera Utara

- Provided daily IT support and troubleshooting for office systems, and conducted monitoring & evaluation (Monev) of ICT infrastructure.
- Designed and delivered a phishing-awareness training session for 40+ staff to help strengthen everyday security habits in the office.
- Built a data visualization dashboard and migrated reporting data to Google Sites, making information easier and faster to access.

Laboratory Teaching Assistant *Sep 2023 – Dec 2025*

Information Technology Laboratory Group, Universitas Sumatera Utara

- Instructed practicum sessions across 5 courses (Web Programming, Data Structures & Algorithms, Data Communications & Computer Networks, Integrative Programming, Network Design & Administration) to 150+ undergraduate IT students.
- Guided a Web Programming student's group to a final project later awarded Top 1 Best Final Project in the IT program at USU (2023/2024).

- Mentored multiple student groups in redesigning a faculty's network architecture for their final projects, successfully guiding them to implement their designs using Software-Defined Networking (SDN).
- Evaluated final projects and collaborated with fellow teaching assistants to develop and improve practicum modules.

PROJECTS

Web Application Penetration Test Report - Merdeka Siber Lab
s.id/PentestReport

- Conducted penetration testing on a simulated web application using Burp Suite as the primary tool, identifying and documenting 8 vulnerabilities including Open Redirect, Username Enumeration, IDOR, and Unrestricted File Upload.
- Exploited vulnerabilities using Proxy, Intruder, and Repeater to intercept requests, brute-force, and tamper with parameters, retrieving system flags as proof of concept.
- Delivered technical mitigation recommendations based on CVSS 3.1 scoring and OWASP/CWE best practices.

ITLG Lab Management System – Backend Engineer
github.com/welmnlg/lab_management_system

- Engineered backend architecture (Laravel, REST APIs) for a laboratory platform, securing APIs with JWT authentication and role-based access control (Admin, Assistant, User).
- Built a QR-based attendance and access system with real-time monitoring, and collaborated on database design and deployment using PostgreSQL.

EDUCATION

Universitas Sumatera Utara, Medan *Jul 2022 – Present*
Bachelor of Information Technology - GPA: 3.85 / 4.00

SMA Negeri 17, Medan *Jun 2019 – Jun 2022*

ORGANIZATIONAL EXPERIENCE

Cybsersecurity Speaker *Jun 2026*
Google Developer Groups on Campus, Universitas Sumatera Utara

- Delivered a hands-on session on Reverse Engineering & Binary Analysis to 70+ participants, covering assembly, static and dynamic analysis, disassembly, and buffer overflow fundamentals using Ghidra and GDB.
- Designed and developed a custom TryHackMe room with CTF-style challenges, enabling participants to apply binary analysis, debugging, and basic exploitation techniques in a practical environment.
- Guided participants through the hands-on challenges using Ghidra and GDB, helping them bridge theoretical concepts with practical reverse engineering and binary exploitation.

Committee Chairperson *Sep 2024 – Dec 2024*
Information Technology Christmas Committee, Universitas Sumatera Utara

- Led a 67-member committee across two student cohorts (2022 & 2023) and 7 divisions, overseeing the Event and Public Relations coordinators.
- Identified and resolved early miscommunication and low engagement among newer members by clarifying role responsibilities and encouraging open communication across seniority levels, resulting in noticeably improved team engagement and task completion.
- Revived and delivered the annual Christmas event after a one-year hiatus for 200+ attendees, while leading fundraising that exceeded the target and raised over Rp31 million within 3 months.

ACHIEVEMENTS

- 3rd Place, Hackathon Siber Tangguh Bangsa Merdeka 2025 (Fakultas Vokasi USU) - solved 30 web exploitation challenges as part of team "Hengker Pesbuk" in a regional Capture the Flag (CTF) Hackathon.
- Awardee, Bakti BCA Scholarship 2024 - Selected from over 1,000 applicants; received educational funding and a one-year self-development program.
- 1st Place, University Business Impact Plan Competition (BCA) - assisted MSME product "Inaklin" in driving sales growth and product awareness.